Heathkit SB-102 Transceiver — Restoration Plan

Rev. 1 — Heath service bulletins, W5RKL, K5BCQ and field sources (manual cross-check pending)

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Safety. The SB-102 runs from an HP-23 supply delivering **~700–800 V DC** to the finals.

That voltage stays on the filter capacitors after the supply is unplugged. Discharge before every hands-in session; never run the final cage open; work one-handed when live. Use a **50 Ω dummy load (not a light bulb)** for all transmit work.

1. Source documents

Document	Where	Use
Heath SB-102 Full Assembly Manual; **SB-102 Alignment & Schematic**	W5RKL (w5rkl.com/manuals)	Primary reference. *Scans only; cross-check pending (§11).*
SB-102 solid-state LMO schematic	archive.org (K6KJN upload)	LMO circuit.
SB-102 Service Bulletins -1 … -30 (1970–84)	Heath list (puck.nether.net), Aug 1997 and Apr 1998	Factory fixes. Many refer to SB-100, SB-101, HW-100 and HW-101 bulletins, also transcribed there.
W5RKL, *Heathkit HW-101 and SB-10x Restoration and Troubleshooting*	w5rkl.com	SB-102 LMO power, electrolytics, bias and loading targets.
K5BCQ SB-101/102 repair site	qsl.net/k5bcq	Leakage limits, IF filter centering, dial mechanics.
Nostalgic Kits Central SB-10x troubleshooting	nostalgickitscentral.com	Measured SB-102 resistor drift and electrolytic leakage.
W2PA SB-101 rebuild notes	sites.google.com/site/radiow2pa	LMO connector, bandswitch rotors, driver neutralizing.
Heath list / Heathkit-Radios / heathkit (groups.io)	puck.nether.net, groups.io	Field experience.

2. Transceiver at a glance

2.1 Key facts

Item	Value
Built	1970–1975, list price $380. Successor to the SB-101; parent design of the HW-101.
Bands	80, 40, 20, 15, 10 m (eight 500 kHz segments, 3.5–29.5 MHz)
Modes	SSB (USB/LSB), CW; optional 400 Hz CW filter
IF	3395 kHz; 2.1 kHz SSB filter; 1st IF 8.395–8.895 MHz bandpass
Output	About 100 W
LMO	**Solid-state** (not TRW), shared with the SB-303/313. Powered at about **+10 V** from the 12.6 VAC filament line **through a #1815 dial lamp**, a 56 Ω resistor, zener **D907**, a **2N3567** regulator and filter caps **C941/C942 (500 µF 25 V)**.
Metering	GRID, ALC, REL PWR, PLATE, HV (the HW-101 lacks GRID and HV)

Item	Value
Extras over SB-101	Front-panel internal/external LMO switch; 2 m transverter (SB-500) socket; headphone and sidetone volume controls on the rear audio board
Supply	**HP-23** family (B+ switch on 300 V)
Boards	Modulator 85-127-3, Audio 85-130-3, Bandpass 85-129-4

2.2 Tube complement

These V numbers are inferred from the SB/HW-10x family and W5RKL. **Verify them against the manual before ordering.**

V	Tube	Function	Known issues (source)
V1	6EA8	Speech amp / cathode follower	GE 6EA8 oscillates → VOX relay chatter (SB-101-19).
V2	6AU6	Carrier isolation amp	—
V3	6AU6	1st IF amp	**Brand-critical.** Heath stocked only GE; non-GE → S-meter drift and IF oscillation (SB-102-30). Sylvania, RCA and Westinghouse → low output and poor VOX (SB-102-24). Leaky → S-meter drift (SB-101-22).
V4	6AU6	2nd IF amp	Same as V3.
V5	6EA8	1st transmit mixer	—
V6	6CB6	2nd transmit mixer	Under shield; shield to socket center pin.
V7	6CL6	Driver	Under shield.
V8, V9	6146 / 6146B	Finals	**Tung-Sol vs. RCA 6146 capacitance changes drive and alignment** (SB-102-3). Re-neutralize after any change.
V10	**6HS6**	Receiver RF amp	Leaky → poor AGC and S-meter decay (SB-102-15 → HW-101-24). Shorts → lost sensitivity and “static” on the S-meter. A 6AU6 substitutes directly.
V11	**6AU6**	1st receive mixer	The SB-102 uses a 6AU6 here; the HW-101 uses a 6HS6.
V12	6EA8	2nd receive mixer / relay amp	A bad tube holds the relays in after transmit (SB-102-14).
V13	6BN8	Product detector / AVC	—
V14	6GW8	Audio amp / output	Hot; C304 sits next to it.
V15	6EA8	CW sidetone osc/amp	—
V16	12AU7	Carrier oscillator	Plate resistors R6/R7 drift (§3.2).
V17	12AT7	VOX amp / 100 kHz calibrator	One restorer found an open filament though it tested good elsewhere.
V18	0A2	150 V regulator	—
V19	12AT7	Heterodyne osc / cathode follower	Repeated failure: R212 220 Ω → 330 Ω (SB-102-2).

That is 19 tubes. Unlike the HW-101, there is **no V20 VFO buffer**; the LMO is self-contained.

2.3 Semiconductors (restoration-kit cross-reference)

Heath p/n	Type	Use	Modern replacement
57-27 (×11)	Silicon rectifier/switching	ALC, recovery, VOX, relays	1N4005–1N4007
FH-110 (×4)	Hot-carrier	Balanced modulator (CR1–CR4)	1N5711
56-26	Germanium	Calibrator / detector / REL PWR	1N34A
56-56	Silicon signal	—	1N4149/1N4148
56-25	15 V zener	—	1N4744A
56-67	10 V zener	LMO supply (D907 area)	1N4740A
—	2N3567	LMO regulator	Any NPN of equal rating; check pinout

Bulletin SB-101-4 covers diode leakage in this family. Leaky **D902/D903** make the ALC meter fall off after 30–60 min. **D101** and **D905** give slow or no receive recovery. **D201** gives poor VOX sensitivity or a long VOX delay. **D907 reversed** gives low sensitivity on 15/10 m (SB-101-10). Bulletin SB-102-28 is a germanium diode change, but its text was not available.

3. What fails, ranked

3.1 Capacitors

All seven electrolytics (W5RKL and restoration-kit lists; one restorer found most of them excessively leaky):

Ref	Value	Role	Replace with
C941, C942	**500 µF 25 V**	**LMO power supply filter** (SB-102 only)	470–500 µF 50 V
C945	10 µF 150 V	Bias filter	10 µF 160–250 V; **negative supply, so check polarity**
C2	10 µF 15 V	Speech amp cathode (modulator board)	10 µF 50 V
C212	10 µF 15 V	VOX amp cathode (bandpass board)	10 µF 50 V
C12	20 µF 350 V	Speech amp B+ decoupling	22 µF 450 V
C304	20 µF 350 V	Audio output decoupling, beside V14	22 µF 450 V. One measured **3.6 mA** leakage (0.5 mA allowed) and would not reform.

Other capacitor	Likelihood	Symptom	Action
.22 µF blue Mylars (×6), C13	Medium	Leakage	Replace with film (kit: 0.22 µF 250–630 V).
C938 (.05 µF)	Medium	—	0.05 µF 630 V film.
C300, C301 (.022 µF, audio board)	Medium	—	0.022 µF 400 V film.

Other capacitor	Likelihood	Symptom	Action
C213 (VOX delay)	Medium	VOX delay drifts with warm-up	Check value and leakage (SB-100-3).
IF transformer internal micas	Low–medium	Corroded → value change or short	Measure; repair the can or replace (K5BCQ).
C14 carrier null trimmer	—	Nulls fully meshed	Move C18 (12 pF) to the other section (SB-102-18).

Leakage limits for paper and film caps (K5BCQ): < 0.01 µF ≤ 0.5 µA; 0.01–1 µF ≤ 1 µA (aged). Electrolytics: I ≤ kC + 0.3 mA, with k = 0.02 (101–250 V) or 0.035 (251–350 V).

3.2 Resistors

Measured drift in one SB-102 (Nostalgic Kits Central; 15 replaced):

Nominal	Measured	Error
100 kΩ (×5)	115 k – **851 k**	+15 to +751%
150 kΩ (×2)	192 k / **open**	+28% / open
330 kΩ	463 k	+40%
22 kΩ	42 k	+92%
3.3 kΩ	5.5 k	+67%
68 kΩ	92 k	+36%

Designators to check first:

Ref	Value	Failure / symptom	Fix (source)
R107	100 kΩ (82 kΩ after SB-101-1)	S-meter zero and drift	100 kΩ 1 W (SB-101-9); check which value is fitted.
R106	22 kΩ 1 W (IF board)	Heat drift → meter zero wander	22 kΩ 2 W wirewound or 33 kΩ 2 W (field).
R6, R7	33 kΩ (carrier osc plates)	Drift → low carrier on one sideband	33 kΩ 1 W (restoration kit).
R116	(kit lists 4.7 kΩ; the description says 47 kΩ)	—	Confirm the value in the manual.
R212	220 Ω	Heterodyne-osc tube failures	330 Ω 1/2 W (SB-102-2).
R117	3.3 kΩ (AVC)	Open → fast AVC/S-meter decay	Replace (via HW-101-55).
R940, R701, R904	Driver / switch boards	Burned from a bandswitch lug shorting to the shield	Replace; bend the lug clear (W5RKL).
56 Ω (LMO supply)	—	LMO voltage low or unstable	Measure.
10 kΩ plate-supply resistors (driver/audio)	—	Drift → low output or weak audio	Measure and replace.

Lift one lead to measure, because parallel paths mislead.

3.3 Tubes, by risk

1. **6AU6 at V3/V4:** brand matters; use GE (SB-102-24, -30).
2. **6HS6 V10:** leakage or shorts; check pin 1 for > +1 V (bad).
3. **6146 pair:** emission, gas, brand-dependent drive.
4. **6EA8 (V1, V12, V15):** oscillation, relay hang, sidetone.
5. **12AT7 (V17, V19):** calibrator and heterodyne oscillator; check filaments.

Use a tester with a **grid-leakage** test (an emission tester misses grid emission). Return each tube to its original socket.

3.4 LMO (the SB-102's weak point)

Symptom	Cause	Fix (source)
LMO dead or intermittent	**#1815 dial lamp open.** The lamp is in series with the LMO supply. Also check D907, the 2N3567, C941/C942 and the 56 Ω resistor.	Replace the lamp (keep a spare); recap C941/C942; check about +10 V regulated (W5RKL).
Instability, jumps	Chassis stress at the mounting lugs	Replace lockwashers and nuts with #6 flat washers and **hex lock nuts**; tighten one turn at a time, criss-cross (SB-102-4).
Jumps when the cable moves	Dirty LMO output phono jack/plug	DeoxIT the collar and shell (SB-101-16; W2PA).
Erratic tuning (early LMOs)	Tuning-cap wiper	Clean the wiper; use a metal washer in place of the fiber washer at the gear assembly (SB-101-12).
Dial correct at 100 kHz points, off by 3–10 kHz between	**Characteristic** of the solid-state LMO (K9STH, five units)	Not a fault. Zero-set on the nearest calibrator point before operating. Bending the slotted rotor plates is possible but tedious.
Stiff tuning, slipping knob	Dial pinch roller and ball vernier	Engage the roller, then tighten the bushing just enough to stop slipping; drive-disk gap 0–0.002 in. (K5BCQ). Raise the knob-shaft bushing to loosen, lower to stiffen (W2PA).
Cracked or crazed dial	Lamp heat	Correct lamp type and shield position; replacement dials exist.

Do **not** fit an SB-102 LMO into an SB-101; it may be unstable (W5RKL).

3.5 Mechanical, switches, relays

Item	Symptom	Fix (source)
Bandswitch rotor/wafer orientation	Dead or partial bands	Check the notches; W2PA found 3 of 4 wrong.
SSB/CW filter slide switch	Low TX **and** low RX	Clean (SB-102-25).
RL2 contact carbon	Chirp, slow recovery	Clean or replace RL2 (SB-102-21).
RL1 power lead	No PTT/TUNE	Check the grey lead (W2PA).
Relay chatter in VOX	GE 6EA8 at V1; noise	SB-102-19; SB-100-3 signal tracing.
Relays stay energized	V12	Replace V12 (SB-102-14).
Comb brackets (aluminum)	Oscillation, low drive on 15/10 m	Steel brackets 204-2096, soldered (SB-102-9); tighten the RF driver board screws and add lockwashers (SB-102-10).
Low 40 m output	Ground-wire routing	Re-route the ground wires; remove 4 spring clips; realign (SB-102-5).
Preselector drive belt and cap	Poor tracking (worst on 10 m)	New belt (silicone 50/56/3 mm); lubricate the bearings; replace cap 26-122 if it binds (SB-102-17).
Loading cap turns with plate	—	Grommet 73-3 (SB-102-20).
Grounds	Hum, instability	Tighten the phono sockets, final socket rivets (drill out → 6-32 if loose) and rails (SB-101-5D, SB-100-4).
Filament wire	Short to a board screw	Re-route the brown/white wire (W2PA).
Lifted foil pads	Open circuits	Secure the pad.

3.6 HP-23 power supply (do first)

Same as for the HW-101:

- Recap the high-voltage section (C1–C4 → 150 µF 500 V).
- Recap C5 (40 µF B+). An open C5 causes a tone on CW and relay chatter (SB-102-29 matches this symptom).
- Recap the bias caps C6/C7 (often open).
- Replace the bleeders (100 kΩ 2 W) and diodes (1N4007).
- Fit a grounded cord.
- **B+ switch on 300 V.**
- Expected outputs: −130 V bias, 300 V B+, about 700–800 V HV, 12.6 VAC filaments.
- **Reform or bring up slowly on a variac** (5–10 min).

4. Restoration plan

4.1 Phase 0: Assess cold

- Photograph everything. Note mods, wrong-value parts (W2PA found four) and non-stock tubes. Record tube brands.
- Pull and label the tubes; test with a grid-leakage tester.
- Check bandswitch rotor orientation.

- Inspect: C304 area, R940/R701, RL1/RL2, belts, the LMO phono cable and the **#1815 lamp**.
- Resistance-check B+, HV, bias and filament lines to chassis against the manual before connecting a supply.

4.2 Phase 1: HP-23

Recap and test the HP-23 unloaded on a variac (§3.6).

4.3 Phase 2: Clean

Vacuum and brush. Use cleaner on swabs, not sprayed. DeoxIT all rotary switches, pots, the **SSB/CW slide switch** and the **internal/external LMO switch**. Keep cleaner off the dial plastic.

4.4 Phase 3: Recap and resistor survey

1. Replace all **7 electrolytics**, including **C941/C942**. Watch C945's polarity.
2. Replace the Mylars and film caps listed in §3.1.
3. Survey resistors by board: audio → modulator → IF → bandpass → RF driver. Replace anything > 10% out (or near the limit), using carbon film.
4. Apply the drift fixes: R106, R107, R6/R7.
5. Replace leaky silicon diodes (SB-101-4 list). Optionally replace FH-110s with 1N5711s.
6. Clean RL1/RL2.

4.5 Phase 4: Mechanical and LMO

Fit the belts; apply the comb-bracket/lockwasher checks; do the LMO mounting fix (SB-102-4); clean the LMO jack; clean the wiper; set the dial roller tension.

4.6 Phase 5: First power-up

1. HP-23 on a variac; dummy load and speaker connected.
2. Confirm the dial lamp is lit and the LMO supply is about +10 V.
3. Log voltages against the manual chart.
4. Calibrator receive check on all bands.
5. **Set bias before any transmit.**

4.7 Phase 6: Alignment

Order: **receiver first.** Warm up 30 min. Use the plastic tool. **Never adjust the sealed 6.8 MHz trap.** Leave T202 alone unless sweep-aligning.

Receiver (shared SB/HW-10x method; verify the manual's test-point values)

1. **S-meter zero:** RF gain fully CW, antenna off. If it won't zero, see R107 (SB-102-1 → SB-101-1/-9).
2. **Heterodyne oscillator coils:** 11 MΩ meter at the bandpass-board test point near V19. Target as high as possible between −0.5 and −2.0 V, **not exceeding −2.0 V**; approach from the slow-changing side.
3. **IF:** peak T201, T102 (top and bottom slugs) and T103 on the calibrator. Keep slugs toward the outside of the cans to avoid overcoupling (K5BCQ).
4. **LMO:**
   - Endpoints 5.000/5.500 MHz.
   - Zero-set the dial.
   - Adjust the LMO frequency shifter for zero beat in LSB vs. USB at the same dial setting.

    - Expect 3–10 kHz error between 100 kHz points (§3.4).
5. **Driver grid/plate coils** in order: 3.5 → 29 → 21 → 14 → 7, preselector matched, peak the S-meter. On 40 m, check SB-102-5 has been done.

Receiver benchmark (W5RKL): calibrator about S9+20 at 3.7 MHz, about S6 at 29.2 MHz.

IF centering check: - Switch LSB ↔ USB on a quiet band. The noise level should match. - Carriers should sit 1.4 kHz each side of the filter center (USB 3396.4, LSB 3393.6 kHz for a 3395.0 kHz center). - To retrim a crystal, 100 pF in series moves it up about 100 Hz; 10 pF in parallel moves it down about 100 Hz.

Transmitter (dummy load; short key-downs; cool down between tries)

1. **Bias:**
    - PTT, no drive, METER on PLATE.
    - Set BIAS ADJUST for about **50 mA** idle (W5RKL).
    - If it reads > 100 mA, hold PTT for only a few seconds.
2. **Grid drive (SB-102-3):**
    - METER on GRID. Advance MIC/CW LEVEL only until grid current just shows, then peak for **maximum grid drive**.
    - Full-scale grid current is normal after tune-up.
    - Roughly 0.3 mA on 3.5 MHz during initial setup (W5RKL).
3. **Carrier isolation (T1)** and **driver neutralizing wire:**
    - Grid current low on some bands with normal output usually means the driver neutralizing wire needs repositioning (W2PA).
    - A wire pushed too deep into hole W makes final neutralizing difficult.
4. **Final neutralization:**
    - Peak REL PWR and plate dip must coincide on FINAL.
    - Alternative (SB-102-8 → SB-100-2): **remove the accessory plug** to kill screen voltage, and disconnect HV (discharge first). Then on 28.5 in TUNE, peak the preselector and FINAL into a dummy load with an RF probe, and null the neutralizing capacitor.
    - If C913 bottoms out, see SB-102-27.
5. **Carrier null:**
    - Equal on USB/LSB; spec −45 dB or better.
    - If the trimmer is fully meshed, see SB-102-18.
6. **Loading:**
    - Dip to about **225–250 mA** at full drive; REL PWR peaks with the dip.
    - Limit key-down to 30 s and cool ≥ 1 min.
7. **Calibrator:** zero-beat against WWV (10/15 MHz) during an unmodulated period.
8. Re-neutralize after any tube change.

4.8 Phase 7: Optional improvements (reversible)

The HW-101/SB-10x improvements apply (WB8JKR, K4JPN):

- 10 MΩ meter-zero resistor for transmit.
- C17 12 → 100 pF and R6/R7 → 27 kΩ for more BFO injection.
- C119 500 pF → .001 µF for less raspy audio.
- C11 .001 → .01 µF.
- R328 470 kΩ → 1 kΩ for faster CW relay pull-in.
- Sidetone bleed fixes.
- .01 µF filament bypass for birdies.
- 200 Ω multi-turn carrier-null pot.
- A replacement solid-state LMO regulator (e.g. LM317-based) is used by some owners. Keep the original if possible.

5. Bulletin quick-reference (symptom → bulletin)

Symptom	SB-102 bulletin (→ referenced)
Meter won't zero / S-meter drift	-1 (→ SB-101-1), -22 (→ HW-100-17), -30
Heterodyne-osc tube failures	-2 (R212)
Low grid drive / alignment	-3, -9, -26
LMO instability	-4; SB-101-12, -16
Low 40 m output	-5
VOX / audio	-6 (→ SB-100-3), -19, -24, -29
Instability / oscillation	-7 (→ SB-100-4), -9, -10
Neutralizing	-8 (→ SB-100-2), -27
Relays hang / chirp	-14, -21
AGC / IF sensitivity	-15, -23, -25
Calibrator spurs	-16
Preselector / loading	-17, -20
Carrier null	-18
Germanium diodes	-28

6. Operation notes

- HP-23 switch on **300 V**.
- Always use a load.
- Zero-set on the nearest 100 kHz calibrator point, because of the LMO's between-point error.
- The internal/external LMO switch must be on INT unless an SB-640 or similar is connected. A wrong setting looks like a dead receiver.

7. Tips and tricks

- Keep spare **#1815** lamps; a dark dial can mean a dead LMO.
- Keep GE 6AU6s for V3/V4, a 6HS6 (or a 6AU6 substitute), and a matched 6146 pair.
- Label tubes and photograph switch rotors before disassembly.
- Measure resistors with one lead lifted.
- Reform electrolytics slowly (5–10 min ramp) if you are keeping any.

8. Maintenance schedule

Interval	Task
Each session	30 min warm-up; calibrator zero-set.
6 months	Exercise switches; check S-meter zero.
Yearly	Bias, carrier null, grid drive; receive benchmark; LMO phono jack clean.
2–3 years	Neutralization check; retighten the driver board and comb screws; DeoxIT.
5 years	Tube test (V3/V4/V10 and finals first); belts; HP-23 check; C941/C942 ESR.
~15–20 years	Re-check electrolytics.

9. Parts sources

- SB-102 restoration kits (e.g. W8AOR; eBay kit sellers): electrolytics including C941/C942, films, diodes, zeners, R6/R7, carrier-null pot.
- Mouser / Digi-Key; HP-23 rebuild kits.
- Heathkit-Radios and heathkit (groups.io): LMOs, crystals, dials, belts.

10. Sources

- Heath, *SB-102 Assembly Manual* and *Alignment & Schematic* (W5RKL); SB-102 LMO schematic (archive.org).
- Heath list (puck.nether.net): SB-102 bulletins (Aug 1997, Apr 1998); SB-101 and SB-100 bulletins (Apr 1998); HW-101 bulletins (Jul 1997).
- W5RKL, *Heathkit HW-101 and SB-10x Restoration and Troubleshooting*.
- K5BCQ, SB-101/102 Repair Site; Nostalgic Kits Central SB-10x troubleshooting; W2PA SB-101 rebuild notes.
- Heath list threads: “SB-102 LMO” (Jun 2002, K9STH), “SB-101 vs 102” (Oct 2009).
- W8AOR SB-102 restoration kit parts list.

11. Pending manual cross-check

Items to verify against the manual:

- V-number assignments.
- The full electrolytic list (C2, C12, C212, C304, C941, C942, C945) and ratings.
- The LMO supply circuit (lamp, 56 Ω, D907, 2N3567).
- Bias and grid-drive procedure and readings.
- Heterodyne test-point values.
- Pilot lamp types.
- R116 value.
- The manual’s alignment and troubleshooting charts.